

Assignment 2 Spring 2025

TTK 4135 Optimization and Control

Adam Godtland Moumen

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1 Problem 1: LP and KKT conditions

1.1 1a

$$\min_x c^T x \text{ s.t. } Ax = b, x \geq 0 \quad (1a)$$

$$c \in \mathbb{R}^n, x \in \mathbb{R}^n, A \in \mathbb{R}^{m \times n} \quad (1b)$$

For x^* to be a solution there exist λ and s s.t. these conditions hold:

$$\mathcal{L}(x, \lambda, s) = c^T x - \lambda^T (Ax - b) - s^T x \quad (2a)$$

$$\nabla_x \mathcal{L} = c - A^T \lambda - s = 0 \Rightarrow A^T \lambda + s = c \quad (2b)$$

$$Ax = b \quad (2c)$$

$$x \geq 0 \quad (2d)$$

$$s \geq 0 \quad (2e)$$

$$x_i s_i = 0 \quad (2f)$$

1.2 1b

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y), \forall \alpha \in [0, 1] \quad (3a)$$

$$f(x) = c^T x \quad (3b)$$

$$c^T(\alpha x + (1 - \alpha)y) \leq \alpha c^T x + (1 - \alpha)c^T y \quad (3c)$$

$$f(\alpha x + (1 - \alpha)y) \leq \alpha f(x) + (1 - \alpha)f(y) \quad (3d)$$

2 Problem 2: Linear programming problem

2.1 2a

Objective function: maximize profit, minimize negative profit. Constraints refer to $x \geq 0$ and the time limits $Ax = b$.

$$f(x) = -c^T x \quad (4a)$$

$$x \in \mathbb{R}^3 \text{ref}[R, S, T]^T \quad (4b)$$

$$c = \begin{bmatrix} 100 \\ 75 \\ 55 \end{bmatrix} \quad (4c)$$

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \quad (4d)$$

$$b = \begin{bmatrix} 7200 \\ 6000 \end{bmatrix} \quad (4e)$$

2.2 2b

Because the A matrix is 2x3 and b is 2x1, we can set one of the x values to zero and keep 2 solveable equations.

$$Ax = b, x = [x_1, x_2, 0]^T \quad (5a)$$

$$\begin{bmatrix} 3 & 2 & 1 \\ 2 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ 0 \end{bmatrix} = \begin{bmatrix} 7200 \\ 6000 \end{bmatrix} \quad (5b)$$

$$\text{Make } A \text{ } 2 \times 2 \text{ and } x \text{ } 2 \times 1 \quad (5c)$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = A^{-1}b \quad (5d)$$

$$x = \begin{bmatrix} 1 & -1 \\ -1 & 3/2 \end{bmatrix} \begin{bmatrix} 7200 \\ 6000 \end{bmatrix} = \begin{bmatrix} 1200 \\ 1800 \end{bmatrix} \quad (5e)$$

This gives basic feasible point $x = [1200, 1800, 0]^T$

Set $x_2 = 0$:

$$\begin{bmatrix} 3 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7200 \\ 6000 \end{bmatrix} \begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3/7 & -1/7 \\ -2/7 & 3/7 \end{bmatrix} \begin{bmatrix} 7200 \\ 6000 \end{bmatrix} \quad (6a)$$

$$\begin{bmatrix} x_1 \\ x_3 \end{bmatrix} = \begin{bmatrix} 15600/7 \\ 3600/7 \end{bmatrix} \quad (6b)$$

The last point is not feasible, as x_3 is not bigger or equal to 0.

$$\begin{bmatrix} 2 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 7200 \\ 6000 \end{bmatrix} \quad (7a)$$

$$\begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 3 & -1 \\ -2 & 2 \end{bmatrix} \begin{bmatrix} 7200 \\ 6000 \end{bmatrix} \quad (7b)$$

$$\begin{bmatrix} x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3900 \\ -600 \end{bmatrix} \quad (7c)$$

2.3 2c

We look at the KKT condition $A^T\lambda + s = -c$ for the first and last point. For the first point, to fullfill the condition $xs = 0, s = [0, 0, s_3]$

$$\begin{bmatrix} 3 & 2 \\ 2 & 2 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ s_3 \end{bmatrix} = - \begin{bmatrix} 100 \\ 75 \\ 55 \end{bmatrix} \quad (8a)$$

$$\begin{bmatrix} 3\lambda_1 + 2\lambda_2 \\ 2\lambda_1 + 2\lambda_2 \\ \lambda_1 + 3\lambda_2 + s_3 \end{bmatrix} = - \begin{bmatrix} 100 \\ 75 \\ 55 \end{bmatrix} \quad (8b)$$

Solving for lambda first:

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = - \begin{bmatrix} 1 & -1 \\ -1 & 3/2 \end{bmatrix} \begin{bmatrix} 100 \\ 75 \end{bmatrix} = \begin{bmatrix} -25 \\ -12.5 \end{bmatrix} \quad (9a)$$

$$s_3 = -55 + 25 + 3 * 12.5 = 7.5 \geq 0 \quad (9b)$$

Because for the other point we get $s_2 \leq 0$, the point with s_3 gives the optimal point. Which is:

$$x^* = \begin{bmatrix} 1200 \\ 1800 \\ 0 \end{bmatrix} \quad (10a)$$

$$-c^T x^* = - \begin{bmatrix} 100 & 75 & 55 \end{bmatrix} \begin{bmatrix} 1200 \\ 1800 \\ 0 \end{bmatrix} = -255000 \text{NOK} \quad (10b)$$

2.4 2d

The dual problem can be written as:

$$\max b^T \lambda \text{ s.t. } A^T \lambda \leq c \quad (11)$$

2.5 2e

$$c^T x^* = (A^T \lambda^* + s)^T x^*, \text{ from } 2b \quad (12a)$$

$$\lambda^{*T} A x^* + s^T x^* = \lambda^{*T} A x^*, \text{ from } 2f \quad (12b)$$

$$\lambda^{*T} A x^* = \lambda^{*T} b = b^T \lambda^*, \text{ from } 2c \quad (12c)$$